

Internship Diary

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Company: Centre of Excellence Digital Forensics Intelligence

Designation: Cybersecurity Internship Program

23 January 2026

Entry:

Introduced to basics of cyber security. connected remote server using ssh. completed till level 4 in over the wire (bandit)

Learnings:

understood basics of cyber security. learnt how to connect remote server using ssh. Learnt basic linux command.

Hours: 4 hours

Total: 4 hours

24 January 2026

Entry:

learnt more about ssh . continued solving over the wire(bandit) challenges and completed till the level 7 using linux commands

Learnings:

learnt more about ssh , learnt about command lines.

Hours: 4 hours

Total: 8 hours

25 January 2026

Entry:

Researched different file system structures and access control mechanisms.

Learnings:

Understood file system architecture.

Hours: 7 hours

Total: 15 hours

26 January 2026

Entry:

Studied common vulnerabilities in file handling and privilege escalation basics.

Learnings:

Learned about file-based attack vectors.

Hours: 7 hours

Total: 22 hours

27 January 2026

Entry:

Installed and configured VMware workstations for creating virtual lab environments . solved over the wire(bandit) till level 10 .

Learnings:

learned about how VMware workstation works . learnt more about linux commands

Hours: 6 hours

Total: 28 hours

28 January 2026

Entry:

Learnt basics of IP addressing , subnetting and VPN concept . Prepared a prd for a portfolio and completed the portfolio. Performed a hands on task using ethernet cable to establish connection and transfer files between system.

Learnings:

Learnt about basics of IP addressing , subnet and VPN . Learnt how to create a prd and a portfolio . Also learnt how to transfer file using ethernet cable by establishing the connection between them.

Hours: 6 hours

Total: 34 hours

29 January 2026

Entry:

Corrected and configured VMware Workstation Pro for lab setup. A file system project was assigned to us , Started with the project and created the prds for the project.

Learnings:

learnt how to setup the VMware workstation pro .Learnt about the structure and purpose of file system project.

Hours: 6 hours

Total: 40 hours

30 January 2026

Entry:

Completed the environmental setup for the project. completed the terminal user interface and graphical user interface of the project.

Learnings:

learnt about how to do the environmental setup . Learnt about the graphical user interface and terminal user interface

Hours: 7 hours

Total: 47 hours

31 January 2026

Entry:

Completed the integration , testing and validation of the project . Added some extra features to the project.

Learnings:

Learnt about how to do the integration and optimization.

Hours: 6 hours

Total: 53 hours

1 February 2026

Entry:

Self-study on file system security and access control lists (ACLs).

Learnings:

Understood permission models in depth.

Hours: 7 hours

Total: 60 hours

2 February 2026

Entry:

Worked on file system project . Added features in graphical user interface and terminal user interface.

Learnings:

Learnt more about the graphical user interface and terminal user interface

Hours: 8 hours

Total: 68 hours

3 February 2026

Entry:

Made changes in gui and tui of the project .Completed the file system project . Prepared the documentation of the file system project. Got introduced to the next project .

Learnings:

Learnt some advanced features necessary for the project .

Learnt about the overview of the next project.

Hours: 7 hours

Total: 75 hours

4 February 2026

Entry:

Started with the shell project. Completed the phase 0 and phase 1 of project .

Learnings:

Learnt about basics of the shell . Learnt about the uses and working of the shell .

Hours: 7 hours

Total: 82 hours

5 February 2026

Entry:

Completed the phase 2 of basic command execution ie to run the basic commands . Completed the phase 3 of shell project argument passing.

Learnings:

Learnt about basics command execution

Learnt about argument passing.

Hours: 7 hours

Total: 89 hours

6 February 2026

Entry:

Completed phase 3 and phase 4 of the shell project. Completed the task of output direction and input direction . Completed the phase 5 command pipelining.

Learnings:

Learnt about the command pipelines. Input output direction.

Hours: 7 hours

Total: 96 hours

7 February 2026

Entry:

Completed the phase 8- Background execution and signal

Learnings:

Learnt about the background execution and signal

Hours: 7 hours

Total: 103 hours

8 February 2026

Entry:

Explored advanced file signature analysis and practiced identifying unknown file types using hex editors.

Learnings:

Improved hex analysis skills.

Hours: 7 hours

Total: 110 hours

9 February 2026

Entry:

Completed the phase 9 of shell project testing cleaning and documentation of the project.

Learnings:

Learnt about how shell project works. Learnt more about the project.

Hours: 7 hours

Total: 117 hours

10 February 2026

Entry:

Make changes in the shell project . Created a tui for the shell project . Solve over the wire (Bandit) till level 12. Completed the ctf .

Learnings:

Learnt more about the linux commands . Learnt about the ctf.

Hours: 7 hours

Total: 124 hours

11 February 2026

Entry:

Learnt about the basic of computer networks. Learnt about the different layers of the computer networks.

Learnings:

Learnt about the computer networks.

Learnt about different layers of network.

Hours: 7 hours

Total: 131 hours

12 February 2026

Entry:

Learnt more about the basics of computer networks

Learnings:

Learnt basics of computer networks

Hours: 7 hours

Total: 138 hours

13 February 2026

Entry:

Completed 3 levels of over the wire Bandit . Learnt about Linux.

Learnings:

Learnt about Linux commands.

Hours: 7 hours

Total: 145 hours

14 February 2026

Entry:

Improved the shell project. Made changes in the tui.

Learnings:

Learnt about Linux commands, projects

Hours: 7 hours

Total: 152 hours

15 February 2026

Entry:

Reviewed virtual lab setup and ensured isolation between different network segments for safe testing.

Learnings:

Learned about network isolation techniques.

Hours: 7 hours

Total: 159 hours

16 February 2026

Entry:

Installed the kali linux . Learnt about the introduction to kali linux and vulnerable machines

Learnings:

Learnt about introduction to kali linux and vulnerable machines

Hours: 7 hours

Total: 166 hours

17 February 2026

Entry:

created a lab setup using Kali Linux and a vulnerable machine (Stapler). configured the network settings so both machines could communicate within the same subnet.

Learnings:

Learnt Network modes

Learnt Subnet and IP addressing concepts

Hours: 7 hours

Total: 173 hours

18 February 2026

Entry:

Tested the connectivity of the kali linux and vulnerable machines using ping commands.

Learnings:

Learnt about ping commands

Hours: 7 hours

Total: 180 hours

19 February 2026

Entry:

started learning basic Linux commands and file system navigation. Practiced commands like ls, cd, mkdir, touch, rm, and learned about permissions and users.

Learnings:

Learnt about File permissions and chmod

Learnt about Root and normal user roles

Hours: 7 hours

Total: 187 hours

20 February 2026

Entry:

performed network scanning on the vulnerable machine using Nmap. Identified open ports and services running on the target system and understood how attackers gather information.

Learnings:

Learnt about Nmap scanning techniques.

Learnt about Open ports and services.

Learnt the difference between TCP and UDP ports

Hours: 7 hours

Total: 194 hours

21 February 2026

Entry:

explored basic web technologies and security concepts. Learned about HTTP, websites, and common vulnerabilities like SQL Injection and XSS at a conceptual level.

Learnings:

Learnt about basics of HTTP and web servers.

Learnt about Common web vulnerabilities.

Hours: 7 hours

Total: 201 hours

22 February 2026

Entry:

Studied asymmetric encryption mechanisms and practiced generating and managing PGP keys.

Learnings:

Understood PKI and key management.

Hours: 7 hours

Total: 208 hours

23 February 2026

Entry:

reviewed all the concepts learned during the internship period. documented my findings and understood how practical labs help in real-world cybersecurity and IT learning.

Learnings:

Practical application of theoretical concepts

Documentation and reporting skills

Hours: 7 hours

Total: 215 hours

24 February 2026

Entry:

performed detailed service enumeration on the target machine after identifying open ports. Analyzed services running on ports like HTTP, SSH, and FTP to understand their versions and possible vulnerabilities.

Learnings:

Service enumeration techniques

Identifying service versions

Importance of banner grabbing

Hours: 7 hours

Total: 222 hours

25 February 2026

Entry:

conducted directory and file brute-forcing on the target web server using tools like Gobuster. I identified hidden directories and explored accessible files on the web application.

Learnings:

Web directory enumeration

Use of wordlists

Importance of hidden directories

Hours: 8 hours

Total: 230 hours

26 February 2026

Entry:

studied password attacks and practiced brute-force attacks in a controlled lab environment. Understood how weak credentials can compromise system security.

Learnings:

1)Types of password attacks

2)Brute-force vs dictionary attack

3)Importance of strong password policies

4)Account lockout mechanisms

Hours: 8 hours

Total: 238 hours

27 February 2026

Entry:

explored privilege escalation concepts. After gaining limited access to the target machine, analyzed system files and permissions to understand how attackers elevate privileges.

Learnings:

1)Concept of privilege escalation

2)Importance of SUID files

3)Checking system misconfigurations

4)Linux permission vulnerabilities

Hours: 9 hours

Total: 247 hours

28 February 2026

Entry:

Focused on basic network security concepts such as firewalls, intrusion detection systems, and packet analysis. Used Wireshark to monitor network traffic in the lab.

Learnings:

1)Packet capturing basics

2)Understanding TCP handshake

3)Role of firewalls

4)Basics of intrusion detection systems

Hours: 9 hours

Total: 256 hours

1 March 2026

Entry:

Researched virtualization networking models (NAT, Bridge, Host-only) and their security implications.

Learnings:

Gained clarity on VM networking modes.

Hours: 7 hours

Total: 263 hours

2 March 2026

Entry:

Studied the penetration testing methodology including reconnaissance, scanning, exploitation, post-exploitation, and privilege escalation. Set up the virtual penetration testing lab environment using Kali Linux and Oracle VM VirtualBox. Configured the vulnerable machine and attacker machine on the same virtual network for testing.

Learnings:

Understood the overall penetration testing lifecycle and the importance of controlled lab environments for practicing ethical hacking techniques.

Hours: 7 hours

Total: 270 hours

3 March 2026

Entry:

Performed network reconnaissance to identify active devices within the lab network. Used the scanning tool Nmap to discover the IP address of the target virtual machine and analyze open ports and running services.

Learnings:

Learned how network reconnaissance helps security professionals identify exposed services and possible attack vectors in a network environment.

Hours: 8 hours

Total: 278 hours

4 March 2026

Entry:

Conducted service enumeration on the target machine to determine the web server and technologies being used. Identified that the web application was running Drupal CMS. Performed directory enumeration to identify accessible files and web resources.

Learnings:

Gained experience in web application reconnaissance and learned how identifying underlying technologies helps in finding potential vulnerabilities.

Hours: 7 hours

Total: 285 hours

5 March 2026

Entry:

Researched publicly known vulnerabilities related to the detected CMS version. Identified a remote code execution vulnerability known as Drupalgeddon2 which affects vulnerable Drupal installations.

Learnings:

Learned how vulnerability research is performed using public vulnerability databases and how outdated software versions can expose systems to exploitation.

Hours: 7 hours

Total: 292 hours

6 March 2026

Entry:

Configured and executed an exploit module using Metasploit Framework targeting the Drupal vulnerability. Successfully established a reverse shell connection to the target machine.

Learnings:

Understood how exploitation frameworks automate vulnerability exploitation and how reverse shells allow remote interaction with compromised systems.

Hours: 7 hours

Total: 299 hours

7 March 2026

Entry:

Performed post-exploitation enumeration after gaining initial access to the target system. Collected system information including user accounts, system configuration files, and directory structures.

Learnings:

Learned the importance of post-exploitation enumeration in identifying further opportunities for privilege escalation and system compromise.

Hours: 7 hours

Total: 306 hours

8 March 2026

Entry:

Reviewed common privilege escalation vectors in Linux and misconfigured system permissions.

Learnings:

Identified Linux security weaknesses.

Hours: 7 hours

Total: 313 hours

9 March 2026

Entry:

Analyzed file permissions and searched for binaries with special permissions in the system. Identified SUID-enabled binaries that could potentially be exploited for privilege escalation.

Learnings:

Developed an understanding of how improper permission settings in Linux systems can lead to privilege escalation vulnerabilities.

Hours: 7 hours

Total: 320 hours

10 March 2026

Entry:

Exploited a vulnerable SUID binary to escalate privileges from a limited user account to the root user. Verified successful privilege escalation and gained administrative access to the system.

Learnings:

Learned how privilege escalation techniques are used to gain full system control after initial access is achieved.

Hours: 7 hours

Total: 327 hours

11 March 2026

Entry:

Started exploring the OSINT CTF platform "OSINT Industries". Understood the objective of Open-Source Intelligence challenges and how publicly available data can be used for investigations. Created an account and familiarized myself with the platform interface.

Learnings:

Learned the basics of OSINT and its importance in cybersecurity investigations. Understood how data footprints are created online and how they can be traced.

Hours: 7 hours

Total: 334 hours

12 March 2026

Entry:

Worked on the "Depix" challenge. Explored techniques to recover hidden or pixelated information from images using OSINT tools.

Learnings:

Learned how pixelation can sometimes be reversed and how image analysis tools can extract sensitive data. Understood risks of sharing blurred screenshots online.

Hours: 7 hours

Total: 341 hours

13 March 2026

Entry:

Solved the challenge "Holehe and the Secret Email". Used email enumeration tools to identify linked accounts and recover hidden email-related information.

Learnings:

Learned how tools like Holehe can check if an email is registered across platforms. Understood account enumeration risks and privacy concerns.

Hours: 7 hours

Total: 348 hours

14 March 2026

Entry:

Worked on "A Taste of Influence – Part 2" challenge. Investigated social media footprints and influence patterns to identify hidden connections.

Learnings:

Learned how social engineering and influencer networks can be analyzed using OSINT. Understood how digital footprints reveal relationships and identities.

Hours: 7 hours

Total: 355 hours

15 March 2026

Entry:

Practiced file analysis techniques and researched steganography basics for hiding data in headers.

Learnings:

Learned digital forensics techniques.

Hours: 7 hours

Total: 362 hours

16 March 2026

Entry:

Revised previously solved challenges and documented step-by-step solutions. Focused on improving investigation methodology and accuracy. Started using Obsidian for documentation. Organized notes related to OSINT tools, techniques, and challenge solutions.

Learnings:

Learned how to use Obsidian for structured note-taking. Understood the importance of maintaining a knowledge base for future reference.

Hours: 7 hours

Total: 369 hours

17 March 2026

Entry:

Installed OPNsense firewall in a virtual environment. Explored initial setup and configuration process.

Learnings:

Learned that OPNsense is an open-source firewall and routing platform. Understood its role in network security and traffic filtering.

Hours: 7 hours

Total: 376 hours

18 March 2026

Entry:

Configured network settings in VirtualBox. Modified subnet and network adapter settings to connect virtual machines.

Learnings:

Learned about subnetting and virtual networking. Understood how multiple VMs communicate within the same network.

Hours: 7 hours

Total: 383 hours

19 March 2026

Entry:

Connected Debian system with OPNsense firewall. Adjusted configurations to establish communication between systems.

Learnings:

Learned how operating systems interact with firewalls. Understood basic network routing and connectivity setup.

Hours: 7 hours

Total: 390 hours

20 March 2026

Entry:

Accessed OPNsense web interface through browser using assigned IP/subnet. Explored dashboard and basic settings.

Learnings:

Learned how firewall interfaces are managed via web UI. Understood monitoring and configuration of network traffic.

Hours: 7 hours

Total: 397 hours

21 March 2026

Entry:

Performed testing of network connectivity between systems. Verified configurations and troubleshooting issues. Consolidated all work done including OSINT challenges, firewall setup, and documentation. Reviewed entire workflow.

Learnings:

Learned basic troubleshooting techniques in networking. Understood importance of IP addressing and correct configuration. Developed a clear understanding of OSINT investigation techniques and basic firewall configuration. Improved practical skills in cybersecurity tools and virtual networking.

Hours: 7 hours

Total: 404 hours

22 March 2026

Entry:

Explored firewall configuration best practices and designed rule sets for multi-tier networks.

Learnings:

Improved firewall management skills.

Hours: 7 hours

Total: 411 hours

23 March 2026

Entry:

Set up the virtual lab environment by accessing the Virtual Network Editor with administrative privileges and understanding how to create isolated network segments. Began configuring custom host-only networks required for the penetration testing lab, focusing on proper planning of IP ranges and segmentation strategy.

Learnings:

Understood the basics of virtual networking and the purpose of network segmentation.

Hours: 8 hours

Total: 419 hours

24 March 2026

Entry:

Created the first custom host-only network (VMnet2) for the attacker machine with the IP range 192.168.10.0/24. Ensured correct subnet configuration and verified that the network is isolated from external internet access.

Learnings:

Learned how to configure an isolated attacker network

Hours: 8 hours

Total: 427 hours

25 March 2026

Entry:

Configured the second host-only network (VMnet3) for target systems using the IP range 192.168.40.0/24. Ensured proper connectivity within the network while maintaining isolation from other segments.

Learnings:

Understood how to set up a target network environment.

Hours: 8 hours

Total: 435 hours

26 March 2026

Entry:

Set up the third host-only network (VMnet4) for management purposes with the IP range 192.168.70.0/24. Focused on structuring this network to later allow controlled internet access.

Learnings:

Learned the role of a management network in a lab setup.

Hours: 8 hours

Total: 443 hours

27 March 2026

Entry:

Configured DHCP settings for all three networks and explored static IP assignment methods. Ensured all systems receive appropriate IP addresses and verified connectivity within each network.

Learnings:

Understood DHCP configuration and static IP assignment.

Hours: 7 hours

Total: 450 hours

28 March 2026

Entry:

Disabled internet access for internal networks (attacker and target networks) to maintain a secure and isolated testing environment. Verified that these networks cannot communicate with external networks.

Learnings:

Learned how to restrict internet access for internal lab networks.

Hours: 7 hours

Total: 457 hours

29 March 2026

Entry:

Self-study on network traffic analysis using Zeek and explored plugins for protocol detection.

Learnings:

Learned event-based monitoring.

Hours: 7 hours

Total: 464 hours

30 March 2026

Entry:

Configured the management network interface to allow internet access while keeping other networks isolated. Ensured proper adapter settings so only the management interface connects externally.

Learnings:

Understood controlled internet access in segmented networks.

Hours: 7 hours

Total: 471 hours

31 March 2026

Entry:

Implemented rules to ensure that the attacker (Red team) network cannot access or ping the management network. Tested connectivity restrictions between networks to confirm segmentation.

Learnings:

Learned how to enforce communication restrictions between networks.

Hours: 7 hours

Total: 478 hours

1 April 2026

Entry:

Performed testing and validation of the entire network setup by checking connectivity between attacker, target, and management networks. Identified and fixed minor configuration issues.

Learnings:

Learned how to enforce communication restrictions between networks.

Hours: 7 hours

Total: 485 hours

2 April 2026

Entry:

Performed testing and validation of the entire network setup by checking connectivity between attacker, target, and management networks. Identified and fixed minor configuration issues.

Learnings:

Gained experience in troubleshooting network configurations.

Hours: 7 hours

Total: 492 hours

3 April 2026

Entry:

Documented the complete virtual lab setup including network configurations, IP schemes, and security restrictions. Reviewed the setup to ensure it meets penetration testing lab requirements.

Learnings:

Learned the importance of documentation in technical projects.

Hours: 7 hours

Total: 499 hours

4 April 2026

Entry:

Introduced to network analysis concepts using Wireshark and understood its role in capturing and analyzing network traffic. Reviewed how packet capture (PCAP) files work and explored the interface of the tool, including filters and packet inspection features to prepare for malware traffic analysis.

Learnings:

Gained basic understanding of Wireshark and packet analysis.

Hours: 7 hours

Total: 506 hours

5 April 2026

Entry:

Analyzed network logs to identify common attack patterns and distinguished normal from malicious traffic.

Learnings:

Improved traffic analysis accuracy.

Hours: 7 hours

Total: 513 hours

6 April 2026

Entry:

Received malware-related PCAP files and began initial analysis on a target machine. Opened and examined the captured traffic in Wireshark, focusing on identifying suspicious packets, unusual protocols, and abnormal communication patterns without executing any files.

Learnings:

Learned how to safely analyze suspicious network data.

Hours: 7 hours

Total: 520 hours

7 April 2026

Entry:

Performed deeper inspection of network traffic by applying filters in Wireshark to isolate specific protocols such as HTTP, DNS, and TCP. Identified possible malicious communication patterns, including unknown IP addresses and unusual request behaviors.

Learnings:

Understood how to filter and analyze specific network activities.

Hours: 7 hours

Total: 527 hours

8 April 2026

Entry:

Followed security precautions while handling potentially harmful files by avoiding execution of any .exe, .sh, or bash files found during analysis. Studied extracted data and payload indicators within the PCAP files to understand malware behavior passively.

Learnings:

Learned safe malware handling practices.

Hours: 7 hours

Total: 534 hours

9 April 2026

Entry:

Configured a secure virtual environment for controlled malware execution testing by enabling core isolation on the Windows host system and disabling clipboard sharing between host and guest machines. Ensured proper isolation before attempting any behavioral observation.

Learnings:

Understood secure environment setup for malware testing.

Hours: 7 hours

Total: 541 hours

10 April 2026

Entry:

Documented findings from the malware traffic analysis, including suspicious IPs, protocols used, and possible attack patterns. Compiled a structured report summarizing observations and insights gained from analyzing the PCAP files.

Learnings:

Learned how to document malware analysis results.

Hours: 7 hours

Total: 548 hours

11 April 2026

Entry:

Started exploring malware traffic analysis resources from Malware Traffic Analysis website and reviewed training exercises to understand how real-world attack data is structured. Focused on understanding how attackers communicate through networks using captured traffic examples.

Learnings:

Understood structure of malware traffic datasets.

Hours: 7 hours

Total: 555 hours

12 April 2026

Entry:

Studied SIEM log normalization and correlation rules in Wazuh for efficient threat hunting.

Learnings:

Understood log processing in SIEM.

Hours: 7 hours

Total: 562 hours

13 April 2026

Entry:

Accessed and reviewed multiple training PCAP exercises from Malware Traffic Analysis and began analyzing sample files on the target machine. Observed different types of network activities and noted important packet-level details.

Learnings:

Learned how to approach malware traffic datasets.

Hours: 7 hours

Total: 569 hours

14 April 2026

Entry:

Analyzed provided PCAP files in detail by identifying communication between systems, focusing on IP addresses, ports, and sessions. Started recognizing normal vs suspicious behavior in network traffic.

Learnings:

Understood traffic flow and communication patterns.

Hours: 7 hours

Total: 576 hours

15 April 2026

Entry:

Studied suspicious traffic patterns by isolating unusual connections and unknown external communications. Compared multiple sessions to identify anomalies that may indicate malicious activity.

Learnings:

Learned to identify abnormal network behavior.

Hours: 7 hours

Total: 583 hours

16 April 2026

Entry:

Followed strict safety practices while analyzing malware-related files by ensuring no executable or script files (.exe, .sh, bash) were run. Focused only on passive analysis to prevent any system compromise.

Learnings:

Understood importance of safe malware handling.

Hours: 7 hours

Total: 590 hours

17 April 2026

Entry:

Referred to video resources to understand real-time malware traffic analysis techniques and how attackers use network communication for command and control. Applied these learnings to better interpret the PCAP data.

Learnings:

Gained insight into attacker communication methods.

Hours: 7 hours

Total: 597 hours

18 April 2026

Entry:

Configured a secure testing setup by enabling core isolation on the host system and disabling clipboard sharing between host and guest machines. Ensured complete isolation before attempting any behavior-based observation.

Learnings:

Learned secure environment configuration.

Hours: 7 hours

Total: 604 hours

19 April 2026

Entry:

Researched adversary emulation profiles in MITRE Caldera and mapped them to the ATT&CK framework.

Learnings:

Mapped threats to MITRE ATT&CK.

Hours: 7 hours

Total: 611 hours

20 April 2026

Entry:

Re-analyzed previously worked PCAP files to validate earlier observations and improve accuracy in identifying suspicious activities. Focused on correlating multiple network events such as repeated connections, unusual request patterns, and consistent communication with specific external IPs to better understand malware behavior.

Learnings:

Improved accuracy in analyzing and correlating network events.

Hours: 7 hours

Total: 618 hours

21 April 2026

Entry:

Started working with adversary emulation tools (Caldera) and explored different attack profiles. Created 15 deepfake images using deepfakesweb.com.

Learnings:

Learned adversary behavior simulation.

Hours: 7 hours

Total: 625 hours

22 April 2026

Entry:

Continued Caldera simulations and monitored system behavior. Generated another 15 deepfake images for the daily target.

Learnings:

Observed automated attack patterns.

Hours: 7 hours

Total: 632 hours

23 April 2026

Entry:

Analyzed Caldera simulation results and documented the findings. Created 15 more deepfake images.

Learnings:

Improved reporting of security simulations.

Hours: 7 hours

Total: 639 hours

24 April 2026

Entry:

Reviewed the integration of various security tools in the lab. Continued with deepfake image generation.

Learnings:

Understood tool interoperability.

Hours: 7 hours

Total: 646 hours

25 April 2026

Entry:

Performed stability tests on the virtual lab environment. Created 15 deepfake images.

Learnings:

Learned environment testing and validation.

Hours: 7 hours

Total: 653 hours

26 April 2026

Entry:

Weekly review and documentation of lab configurations. Explored Burp Suite and PortSwigger Web Security Academy. Created 10 deepfake images.

Learnings:

Started advanced web security training.

Hours: 7 hours

Total: 660 hours

27 April 2026

Entry:

Studied OWASP Top 10 vulnerabilities in detail, focusing on Broken Access Control and Cryptographic Failures. Created 15 deepfake images (Total: 100).

Learnings:

Deepened understanding of web vulnerabilities.

Hours: 7 hours

Total: 667 hours

28 April 2026

Entry:

Explored Burp Suite tools like Repeater and Intruder for web vulnerability testing. Review of OWASP Top 10 Injection and Insecure Design.

Learnings:

Learned to use Burp Suite for testing.

Hours: 7 hours

Total: 674 hours

29 April 2026

Entry:

Started connecting Wazuh and OPNsense for centralized log management. Configured log forwarding from the firewall.

Learnings:

Learned SIEM-Firewall integration.

Hours: 7 hours

Total: 681 hours

30 April 2026

Entry:

Continued Wazuh-OPNsense integration and verified log ingestion. Tested alerts for firewall rule violations.

Learnings:

Validated log collection and alerting.

Hours: 7 hours

Total: 688 hours

1 May 2026

Entry:

Installed Metasploitable2 on the target subnet and verified connectivity. Started setup of OWASP Juice Shop on the same machine.

Learnings:

Learned to deploy vulnerable systems for testing.

Hours: 7 hours

Total: 695 hours

2 May 2026

Entry:

Completed OWASP Juice Shop installation and successfully installed Wazuh and Caldera agents on the target machine. Verified agent reporting.

Learnings:

Successfully integrated agents on target systems.

Hours: 7 hours

Total: 702 hours